Analysis of PowerWorld Simulator Time Step Simulation and Auxiliary Data Block Architecture

The analysis of modern electrical power grids over extended chronological horizons requires simulation engines capable of transitioning between steady-state power flows, optimal dispatches, and transient stability assessments. Within PowerWorld Simulator, the Time Step Simulation (TSS) tool serves as the primary engine for executing sequential chronological simulations[1]. This technical report examines the operational paradigms, algorithmic execution sequences, advanced control mechanisms, and the auxiliary data block syntax that govern the input, execution, and output operations of the TSS tool.

Operational Modes of the Time Step Simulation Engine

The Time Step Simulation framework solves sequential power flow cases at distinct chronological intervals[3]. Time points can be specified at resolutions as fine as one second, with each point linked directly to its associated input parameters and solution results within a structured Timepoint Record[4]. The list of time points is maintained chronologically, and inserting an intermediate date or time causes the engine to automatically sort the list to preserve chronological continuity[4]. If a time point is deleted, its entire associated input and output data record is purged from memory[4].

The TSS tool functions under two distinct execution paradigms, configured on the Options page of the Time Step Simulation dialog[1]:

- **Continuous Simulation:** Designed for rapid analytical computational throughput[1]. The simulation engine solves each chronological time point immediately after the completion of the preceding step[1]. The primary objective is to complete the entire chronological horizon as quickly as the system's CPU resources allow, updating the user interface dynamically to show the evolution of system states[1].
- **Timed Simulation:** Configured to progress proportionally to a real-world clock[1]. The interval between consecutive time points dictates a proportional delay before the solver initiates the next step[1].

The scaling relationship of a Timed Simulation is defined mathematically as:

$$t_{\text{execution}} = \frac{\Delta t_{\text{sim}}}{\text{Time Scale}}$$

In this formulation, Δt_{sim} represents the chronological gap between consecutive time points in seconds, the Time Scale is a dimensionless speed multiplier, and $t_{\text{execution}}$ is the

corresponding wall-clock execution time in seconds[1]. For example, a Time Scale of 60 indicates that a one-hour chronological gap between simulated states will execute over exactly 60 seconds of wall-clock time[1]. This mode supports flow animation, dynamic one-line contouring, and manual operator intervention (such as generator adjustments or capacitor switching) between solved states, making it highly valuable for real-time dispatch training and dynamic system visualization[1].
Regardless of the selected execution mode, the simulation progresses through three discrete operating states[3]:

Simulation State	Engine Behavior and Constraints	Available Operations
Reset	The default initial state or the state entered upon completion or explicit manual reset. Chronological progression is halted; all process flags are set to default[3].	Time point selection, data import, initialization[3].
Running	Sequential chronological execution is active[3]. The engine processes time points starting from the specified Starting Time to the Ending Time[3].	Manual pause trigger, dynamic data monitoring[3].
Paused	Chronological progression is temporarily suspended[3]. The solver always completes the current time point's execution entirely—including full contingency analysis and SCOPF optimization loops—before entering this state[3].	Single-step execution, reverse stepping, parameter modification[3].

The user interface provides interactive controls such as the Do Run button to initialize or resume continuous solving, Do Single Point to step forward a single interval, Do Previous Point to step backward, and Reset Run to re-initialize the processed fields to their default state without deleting calculated results[3].

Chronological Data Input and Hierarchical Scaling

The foundational dataset for a TSS run is entered via the Input page, which utilizes tabular matrix grids to define time-varying parameters[2]. These grids accommodate MW/Mvar loads, generator outputs, line statuses, and custom input variables[6].
To simulate realistic system-wide load variations when complete individual bus-level chronological load datasets are unavailable, PowerWorld Simulator implements a hierarchical load-scaling architecture[5]. Users can define chronological variations at the Area or Zone level[5]. The scaling logic processes load variations through a multi-tiered calculation: first adjusting individual loads to their specified timepoint-specific values, and then scaling the remaining unmapped loads in the Area or Zone to meet the target total[5].
During the scaling phase, reactive power (Mvar) adjustments can be handled using two distinct mathematical methods[5]:

1. **Constant Power Factor:** The reactive power is scaled proportionally with active power to maintain a constant power factor[5].
2. **Fixed Reactive Power:** The reactive load remains locked at its base case value while the active power scales[5].

For aggregate injections that involve both load and generation, **Injection Groups** distribute total chronological MW adjustments based on pre-defined Participation Factors[5]:

$$P_i = P_{i,\text{base}} + \Delta P_{\text{total}} \cdot \left(\frac{\text{ParFac}_i}{\sum_{j \in \mathcal{I}} \text{ParFac}_j} \right)$$

In this equation, $\mathcal{I}$ is the set of online participation points, ParFac_i is the participation factor of the specific point, and ΔP_{total} is the total active power change applied to the injection group[8]. If an injection group's calculated target is zero, the engine bypasses adjustments for that step[6]. To force a true zero injection, a minimal non-zero value (e.g., 0.01 MW) must be entered[6].
Importantly, **Bus Participation Points** are strictly ignored by tools that alter physical system states (such as TSS, scaling, and AGC controls), though they are allowed in linear sensitivity calculations[8]. When saving injection groups to an auxiliary file, if the auto-calculate method is configured as GEN MAX INC and loaded into a different case, the engine automatically recalculates participation factors based on the positive active power reserves of that specific system[8].

Algorithmic Execution Sequence of a Time Step

Each time point processed during a simulation run follows a highly coordinated mathematical and logical sequence to ensure physical validity, proper action of control systems, and accurate data logging[9]. The following sequence is strictly adhered to by the execution engine:

Sequence Phase	Action Subtype	Execution Details and Logical Controls
Phase 1	Pre-Script Execution	Runs user-defined script commands prior to any data changes[9].
Phase 2	Input Data Application	Updates load active/reactive powers, generator output limits, line operational statuses, and Area/Zone load scaling targets[5].
Phase 3	Custom Inputs	Evaluates custom algebraic and mapping inputs[9].
Phase 4	Schedule Application	Integrates chronological Scheduled Transactions and Action Group statuses[6].
Phase 5	Intermediate Scripting	Executes scripts defined to run immediately post-data application[9].
Phase 6	Steady-State Solve	Initiates numerical power flow engine. If non-convergence occurs and "Pause if Power Flow Does Not Solve" is active, the run halts[5].
Phase 7	Control Delay Verification	Evaluates delay timers for transformers, switched shunts, and Time Step Actions[9].
Phase 8	Secondary Solve Loop	If control delay evaluations trigger a device state

		change, the power flow solver executes again to achieve a new steady-state equilibrium[9].
Phase 9	Advanced Optimization	Runs active Optimal Power Flow (OPF), Security-Constrained OPF (SCOPF), or OPF Reserves modules[9].
Phase 10	Contingency Analysis	Executes contingency screening loops if configured[9].
Phase 11	Post-Script Execution	Runs scripting blocks immediately prior to state logging[9].
Phase 12	Results Logging	Stores selected state variables to RAM or direct disk exports[5].

This chronological execution framework also integrates with advanced optimal power flow tools, such as the OPF Reserves module[10]. This integration allows users to run hourly dispatch optimizations containing multi-tiered reserve bids (regulation, spinning, and supplemental reserves) across control areas and zones, storing chronological dispatches, prices, and reserve margin metrics over the entire simulation window[10].

Advanced Control Actions and Delayed Response Modeling

To prevent numerical hunting and accurately simulate physical equipment response times, the TSS engine features specialized time-delay controllers for transformer taps, switched shunts, and conditional logic blocks[9].

Device Controller Delays

Physical voltage-regulating equipment does not react instantaneously to out-of-band voltages. The TSS engine models these switching delays by disabling automatic device control during the initial power flow execution[9]. After finding an initial solution, the simulator evaluates how long the regulating bus has been outside its deadband[9]. Only after the specified time delay is met or exceeded will a control adjustment be implemented in a subsequent iteration[9].

Time Step Actions

These are user-defined conditional events analogous to contingency actions but triggered by steady-state system conditions over a sustained period[11]. Every Time Step Action requires:

- A **Model Criteria** expression that must evaluate to true[11].
- A **Time Delay** defined in whole seconds (minimum value of 1 second)[11].

When the model criteria evaluates to true, an internal timer is initialized[11]. If the criteria becomes false at any subsequent step, the timer is immediately reset to zero[11]. If the timer accumulates a value equal to or greater than the specified Time Delay, the action (such as opening a transmission branch, adjusting load, or scaling generation) is executed[9]. Once a Time Step Action is executed, it is locked out and cannot be triggered again during that run[11].

Both device delays and Time Step Actions are strictly dependent on the simulation running as a **Complete Run**[9]. A Complete Run is initialized via the Do Run button and progresses chronologically without out-of-order stepping or repeated intervals[9]. If a user executes manual single-stepping (Do Single Point) or steps backward (Do Previous Point), all time-delay calculations are bypassed, transformers/shunts resort to standard instantaneous controls, and conditional Time Step Actions are completely ignored[9].

Scheduled Actions and Timeline Animations

Scheduled Actions represent another distinct temporal control type, designed to model discrete, planned schedule changes rather than reactive, threshold-based ones[12]. Configured inside the Scheduled Actions Dialog, these elements are organized into Scheduled Action Groups, Scheduled Actions, and Scheduled Action Statuses[12]. Active schedules are visualized using an interactive Gantt chart, showing the chronological overlap of action groups[12].

The user can define the timeline of interest using a Start Time, End Time, and temporal Resolution, and then animate the scheduled actions sequentially[12]. This chronological animation steps forward through time, applying active schedules at each step, and allows users to save case snapshots at critical time points to facilitate off-line diagnostic assessments[12].

Auxiliary Data Block Architecture and Syntactical Design

PowerWorld Simulator uses text-based Auxiliary Files (.aux) to exchange data, define formatting, and execute batch script actions[13]. These files are divided into structural DATA and SCRIPT sections[13].

Concise Format and Object Identifier Syntaxes

Modern auxiliary files employ concise headers, compact variable names, and the ObjectID field to minimize file size and processing overhead[13]. The parser supports key-field lookup based on standard index formats, names, or user-defined labels[15].

An auxiliary data section starts with the object type name, followed by a list of fields in

parentheses, and the data list enclosed in braces[14]:
Gen (ObjectID, CTGPreventAGC, CTGPartFact, CTGMaxResp, UseLineDrop, Xcomp) { "Gen 'Texan_69.0' '1'" "NO" same 22.0 "NO" 0.0001 }
Legacy files may still use the DATA keyword prefixed format: DATA DataName(object_type, [list_of_fields])[14].
To facilitate dynamic, context-sensitive scripting, the simulator supports special character syntaxes within auxiliary files:

- **Object and Field Referencing (& Character):** Allows a data block to dynamically query a field value from another object[15]. The syntax is structured as: "&objecttype 'key1' 'key2' variablename"
- **Variable Substitution (@ Character):** Resolves values at execution time by querying model options or specific fields from the target record[16].

An example of combining these syntaxes to dynamically scale a load parameter based on zone information is:
SetData(Load, [CustomFloat], ["&Zone '@ZoneNumber' LoadMW"], All)
In this command, the engine dynamically extracts the @ZoneNumber of each load, looks up the corresponding Zone object, retrieves its LoadMW value, and writes that value to the load's CustomFloat field[17]. Furthermore, script parameters can dynamically generate paths and filenames based on system attributes using the @MODELFIELD keyword, which extracts values using the syntax:
@MODELFIELD<objecttype 'key1' 'key2' 'key3' variablename:digits:rod>[18].

Data Representation and Options Formatting

Within auxiliary data files, structural options determine how object attributes are created and updated:

- **DATA vs SUBDATA Sections:** Data records can be written as top-level DATA blocks or as nested SUBDATA blocks associated with a parent container[19]. While older software versions default to writing nested SUBDATA configurations, Version 22 introduced "Save As Auxiliary with Options," which defaults to saving records as flat DATA blocks[19]. This format provides greater parsing speed and flexibility by letting users choose which contained objects to include[19].
- **Auto-Creation Control (AUXDEF, YES):** When exporting tool options (such as Contingency Analysis, Transient Stability, or Limit Monitoring settings) to an auxiliary file, the parser automatically appends , AUXDEF, YES to the section headers[15]. This ensures that when the auxiliary file is loaded, the corresponding option objects are automatically created if they do not exist, eliminating modal dialog prompts that interrupt batch processing[15].

Case Export Formatting and File Interoperability

To manage complex system studies, PowerWorld Simulator provides tools that define file structures, control formatting, and group related files into portable packages.

Complete Case Export Format Descriptions

To ensure consistency when sharing model data, Simulator provides structured format descriptions for complete cases[20]. This utility structures data into logically distinct sections, providing a clear division between static network models and dynamic or simulation-specific parameters[20]:

Export Category	Input Structures and Hard-Coded Definitions
Custom Info	Custom field descriptions, advanced filters, logic conditions, expressions, and calculated fields[20].
Network Model	Standard structural model definitions containing buses, generators, loads, and lines[20].
Contingency	Contingency options, auto-insertion criteria, global actions, remedial actions, and exception monitors[20].
Transient	Transient stability options, plot definitions, limit monitors, and RAM storage variables[20].
Scheduled Actions	Planned action items, scheduled outages, and group temporal dependencies[16].
Time Step	Dedicated inputs and configuration structures for the Time Step Simulation tool[20].

Users can also choose to split the exported dataset into two distinct blocks[21]:

1. **Static Block:** Fields that rarely change (e.g., nominal voltages, physical line impedance)[21].
2. **Dynamic Block:** Fields that change frequently during operational studies (e.g., active power dispatches, tap positions)[21].

File Access Robustness and Tolerant Parsing

Chronological simulations frequently run on multi-core environments where file sharing or lock conflicts can occur. To prevent execution crashes during batch processing, the auxiliary script writing engine features an automatic retry loop[21]. If an output file is locked by another process, the script engine pauses and retries writing up to 100 times with an intermediate delay of 0.1 seconds before aborting the run[21].

Additionally, the parser incorporates tolerant parsing rules to handle syntax variations from external applications[21]:

- **Legacy PSLF Formats:** Old PSLF WECC Remedial Action Scheme (RAS) formats often write the keywords SECDD or TRAN where a standard network model branch is expected[21]. The simulator reinterprets these keywords as BRANCH elements, reading the files successfully while writing them back out in the correct syntax[21].
- **Unlinked Action Handling:** If an auxiliary file contains a contingency or remedial action that references a missing device (such as a decommissioned generator or line), the parser creates an "Unlinked" contingency element[17]. This allows the user to inspect and repair the unlinked action in the user interface rather than halting case loading[17].
- **Format Extensions (!= Operators):** The parser supports alternate logical comparison operators such as != (reinterpreting it internally as <>) to ease integration with standard scripting tools[19].

Project Container Files (*.pwp)

To solve the portability challenges of managing multiple interdependent files during large-scale simulations, PowerWorld provides Project Files with the *.pwp extension[22]. A Project file acts as a structured ZIP-compatible container that packages[22]:

- The primary network model case file (*.pwb)[22].
- One or more one-line diagrams (*.pwd) containing interactive visualization layouts[22].
- One or more text-based case auxiliary files (*.aux) and display auxiliary files (*.axd)[13].
- Batch execution scripts and case templates containing solution and display options[22].

This containerized approach allows users to transfer complete chronological studies between systems in a single step, with Simulator handling file compression, extraction, and script parsing automatically[22].

Result Storage, Output Management, and Data Ingestion

Chronological simulations generate massive datasets, making efficient result storage and data visualization essential for practical analysis.

Comparison of TSS Output Formats

PowerWorld Simulator offers several output options to direct results to computer RAM, local disk, or external databases:

Format Designation	File Extension	Structural Attributes	Primary Operational Use Case
Time Series Binary	.tsb	Proprietary	Saving complete

		compressed binary structure containing inputs, options, schedules, and custom results[2].	TSS states to allow users to pause, reload, and review studies inside Simulator[2].
Auxiliary Data	.aux	Text format utilizing space-delimited fields and standard DATA/SUBDATA block structures[13].	Re-importing results as inputs for other cases or automating workflows via batch scripts[13].
Standard CSV	.csv	Comma-separated variable fields. The initial rows contain the object type and concise field variable headers[14].	Import into external programming environments (such as Python or MATLAB) for post-processing.
CSV with Column Headers	.csv	Comma-separated variable fields. The first row contains the exact column headers used in Case Information Displays[14].	Generating human-readable reports and performing spreadsheet-based manual audits.
CSV No Header	.csv	Comma-separated variable fields containing raw data arrays with no header rows[14].	High-throughput appending operations and ingestion by relational database structures.

These exports can be directed to memory, saved directly to disk as CSV files to prevent RAM exhaustion, or written to both[5].

GIS Contouring and Weather-Dependent Ingestion

To help identify operational issues, the TSS engine integrates with spatial analysis and visualization tools:

- **Automatic Contour Generation:** The simulator can generate one-line contour diagrams at each solved time point to show changing operating conditions (such as voltage profiles or line loadings) over time[1]. These contours can be saved automatically as sequential JPEG or bitmap images, providing a frame-by-frame visualization of system changes[1]. Modern versions also support contouring spatial data without explicit one-line objects, which is highly useful for visualizing geomagnetic induced currents (GIC)[19].

- **Time-Varying Weather Integration:** Advanced simulations can import historical weather data (such as temperature, wind speed, and solar irradiance) represented in the GeoJSON format (*.json)[16]. This weather data is mapped directly to the time-step simulation[16]. At each step, the engine dynamically calculates weather-dependent active power output limits for generators and thermal limits for transmission branches, providing a highly accurate representation of physical system capacities under varying environmental conditions[16].

Technical Recommendations for Multi-Interval Planning Studies

Based on the capabilities of the Time Step Simulation engine and the auxiliary data block structure, several recommendations are provided to help engineers design efficient multi-interval planning studies:

1. **Memory Management:** For chronological simulations spanning more than 168 intervals (one week of hourly data) on networks larger than 5,000 buses, engineers should avoid RAM-only result storage[5]. Direct CSV writing with the CSVNOHEADER format minimizes processing overhead and prevents out-of-memory errors[5].
2. **Maintaining Control System Realism:** When modeling regulating transformers or switched shunts with time-delay characteristics, simulations must be run as a "Complete Run"[9]. Using manual single-stepping or stepping backward during diagnostic assessments disables these delay timers, reverting the devices to instantaneous control and potentially misrepresenting system voltage control behavior[9].
3. **Reducing Redundancy in Auxiliary Files:** When writing custom script tasks, engineers should utilize the concise header option and the dynamic @ and & formatting syntaxes to build generic data queries[13]. Dynamic referencing allows options and controls to adapt automatically to different case files, eliminating the need to maintain separate static data blocks for every individual network model.
4. **Scheduled Action Conflicts:** In complex chronological studies involving numerous planned maintenance events and switching actions, engineers should check for scheduled action conflicts using the built-in conflict checking utilities[12]. This tool scans active schedules and highlights devices that have been assigned contradictory commands (e.g., simultaneous open and close actions), ensuring that schedule configurations are physically logical before running the simulation[12].

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